

Trainer Debrief — Lab 31

Consolidating Your Capstone — The A3 Storyboard

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Consolidate all DMAIC artifacts from Labs 1-30 into a single A3 storyboard (A1-A5, K1).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-31-a3-storyboard.xlsx

The A3 discipline is subtraction, not addition. Thirty labs produced far more than one page holds, so the register exists to decide what goes on the A3 and what goes in the appendix pack you bring to the room but do not present. Panel 2 must carry the baseline Ppk of 0.62 and panel 7 the improved 1.40, measured the SAME way — the single most common capstone failure is a before/after comparison using two different measurement definitions, which the steering committee will spot immediately.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your A3 fits on one page, covers all seven sections, carries your real project data throughout, and a peer reading it cold can correctly summarise the project.

Where this sits in the capstone

You have been building this project since Lab 1. Now assemble thirty labs' worth of artifacts into ONE page that tells the whole story. The A3 discipline is deliberately constraining — if it does not fit on one page, you do not yet understand which parts matter.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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